Power Series Distribution

1 Definition

Now, consider the power series $\sum_x a(x)\theta^x$, let, $\sum a(x)\theta^x$ be convergent for $-p\theta < p, \sum_x a(x)\theta^x = g(\theta)$ for $0 \leqslant \theta < p$.

$$\sum_x \frac{a(x)\theta^x}{g(\theta)} = 1 \quad \text{and also} \quad \frac{a(x)\theta^x}{g(\theta)} \geqslant 0 \quad \forall x = 0, 1, 2, \ldots.$$

Hence, $\left\{ \frac{a(a)\theta^x}{g(\theta)}, x = 0, 1, 2, \ldots \right\}$ describes a PMF and the probability distribution is known as Power series.

1. Set $g(x) = \frac{1}{x!}, x = 0, 1, 2, \ldots$

$\sum_{x=0}^{\infty} \frac{\theta^x}{x!} = e^{\theta} \Rightarrow f(x) = \frac{e^{-\theta}\theta^x}{x!}$. It is a Poisson Distribution.

2. Set $a(x) = \begin{cases} \binom{n}{x} & x = 0, 1, 2, \ldots \\ 0 & \text{otherwise} \end{cases}$,

$$\sum_{x=0}^{\infty} a(x)\theta^x = \sum_{x=0}^{\infty} \binom{n}{x}\theta^x = (1+\theta)^n$$

$\therefore f(x) = \binom{n}{x}\frac{\theta^x}{(1+\theta)^n} = \binom{n}{x}\left(\frac{\theta}{1+\theta}\right)^x \left(1 - \frac{\theta}{1+\theta}\right)^{n-x} <$ It is a Binomial Distribution.

3. Set $a(x) = \binom{x+r-1}{x}, x = 0, 1, 2, \ldots$

$$\sum_x a(x)\theta^x = \sum_x \binom{x+r-1}{x}\theta^x = (1-\theta)^{-r}$$

$f(x) = \binom{x+r-1}{x}(1-\theta)^r\theta^x$ It is a Negative Binomial Distribution.

4. Set $a(x) = 1, \quad \sum_{x=0}^{\infty} a(x)\theta^x = (1-\theta)^{-1}$ for $|\theta| < 1 \Rightarrow -1 < \theta < 1$

$f(x) = (1-\theta)\theta^x, x = 0, 1, 2, \ldots$ It is a Geometric Distribution.

- It is important to note that the power series distribution includes Binomial, Poisson, Negative Binomial, Geometric distributions etc.

2 Mean and Variance

Here, $g(\theta) = \sum_{x=0}^{\infty} a(x)\theta^x, 0 < \theta < p$.

Differentiating w.r.t $\theta, g'(\theta) = \sum_x a(x)x\theta^{x-1}$

Again differentiating w.r.t $\theta, g''(\theta) = \sum_x a(x)x(x-1)\theta^{x-2}$

$$E(X) = \sum_x x\frac{a(x)\theta^x}{g(\theta)} = \frac{\theta}{g(\theta)}\sum_x xa(x)\theta^{x-1} = \frac{\theta}{g(\theta)}g'(\theta) = \theta\frac{d}{d\theta}[\ln g(\theta)]$$

$$E(X(X-1)) = \sum_x x(x-1)\frac{a(x)\theta^x}{g(\theta)} = \frac{\theta^2}{g(\theta)}\sum_x a(x)x(x-1)\theta^{x-2} = \frac{\theta^2}{g(\theta)}g''(\theta)$$

$$\begin{aligned}
&\text{Var}(x) = E(X(X-1)) + E(X) - E^2(X) \\
&= \frac{\theta^2}{g(\theta)} g''(\theta) - \left(\frac{\theta g'(\theta)}{g(\theta)}\right)^2 + E(X) \\
&= \frac{\theta^2 g''(\theta) g(\theta) - \theta^2 \left(g'(\theta)\right)^2}{(g(\theta))^2} + E(X) \\
&= \theta^2 \frac{d^2}{d\theta^2} [\ln g(\theta)] + [E(X)]
\end{aligned}$$

1. A sum related to concept discussed over here. If $f(x)f'(x) < 0, \forall x \in R$
(1) $f(x)$ is monotonically increasing.
(2) $f(x)$ is monotonically decreasing.
(3) $|f(x)|$ is monotonically increasing.
(4) $|f(x)|$ is monotonically decreasing.

$$\begin{aligned}
&f(x)f'(x) < 0. \\
\Rightarrow &\frac{f(x)f'(x)}{f(x)f(x)} < 0 \Rightarrow \frac{f'(x)}{f(x)} < 0 \Rightarrow \frac{d}{dx}(\ln |f(x)|) < 0.
\end{aligned}$$

$\therefore \quad lnf(x) \mid$ is decreasing.

$$\Rightarrow |f(x)| \text{ is decreasing.}$$

2. In a class of power series distribution mean=variance iff the distribution is Poisson.

If Part

Let X be a random variable having a power series distribution with pmf is given by $f(x) = \begin{cases} \frac{a(x)\theta^x}{g(\theta)} & x = 1, 2, \ldots \\ 0, & \text{ow} \end{cases}$

$$g(\theta) = \sum_x a(x)\theta^x, 0 < \theta < p$$

$$\begin{aligned}
&\text{Now, } E(X) = V(X) = \theta^2 \frac{d^2}{d\theta^2} ln(g(\theta)) + E(X) \\
&\Rightarrow E(X) = \frac{d^2}{d\theta^2} \ln(g(\theta)) = 0 \\
&\Rightarrow \ln(g(\theta)) = k\theta + m \Rightarrow g(\theta) = e^m e^{k\theta} = ce^{k\theta} \\
&\Rightarrow c = e^m \\
&\sum_x a(x)\theta^x = c \sum_{x=0}^{\infty} \frac{(k\theta)^x}{x!} \forall \theta \in (0, p)
\end{aligned}$$

Equating the efficient of $\theta^x, \quad a(x) = c\frac{k^x}{x!}$

$$f(x) = \frac{c\frac{k^x}{x!}\theta^x}{ce^{k\theta}} = e^{-k\theta}\frac{(k\theta)^x}{x!}$$

So, $X \sim P(\lambda = k\theta)$

Only if Part

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\begin{aligned}
&\text{If } X \sim \text{ Poisson } (\theta) \\
&g(\theta) = \sum_x a(x)\theta^x = \sum_x \frac{1}{x!}\theta^x = e^\theta \\
&E(X) = \theta \frac{d}{d\theta}\left(\ln e^\theta\right) = \theta \frac{1}{e^\theta} e^\theta = \theta \\
&V(X) = \theta^2 \frac{d^2}{d\theta^2}\left[\ln e^\theta\right] + \theta = \theta^2 \frac{d}{d\theta}\left[\frac{1}{e^\theta} e^\theta\right] + \theta = \theta^2 \cdot 0 + \theta = \theta \\
&\therefore E(X) = \text{Var}(X)
\end{aligned}
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3 Recursion relation for central moment

Kth odder central moment of a R.V X is $E(X - E(X)^k = \mu_k$, If $X \sim$ power Series Distribution, $f(x) = \frac{a(x)\theta^x}{g(\theta)}, x = 0, 1, 2, \ldots$

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\begin{aligned}
a(x) \geqslant & 0, \forall x \text{ and } \sum a(x) = 1 \\
E(X) = & \theta \frac{d}{d\theta}[\ln g(\theta)] \\
\mu_k = & \sum_{x=0}^{\infty} \left[x - \theta \frac{d}{d\theta} \ln g(\theta)\right]^k \frac{a(x)\theta^x}{g(\theta)} \\
\frac{d\mu_k}{d\theta} = & \sum_{x=0}^{\infty} k \left(x - \theta \frac{d}{d\theta} \ln g(\theta)\right)^{k-1} (-1) \left\{\theta \frac{d^2}{d\theta^2} \ln g(\theta) + \frac{d}{d\theta} \ln g(\theta)\right\} \cdot \frac{a(x)\theta^x}{g(\theta)} \\
& + \sum_{x=0}^{\infty} \left(x - \theta \frac{d}{d\theta} \ln g(\theta)\right)^k a(x) \left\{\frac{g(\theta)\theta^{x-1}x - \theta^x g'(\theta)}{g^2(\theta)}\right\}
\end{aligned}
$$

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\begin{aligned}
\Rightarrow \frac{d\mu_k}{d\theta} = & (-k) \left\{\theta \frac{d^2}{d\theta^2} \ln g(\theta) + \frac{d}{d\theta} \ln g(\theta)\right\} \mu_{k-1} + \sum_{x=0}^{0} \left(x - \theta \frac{d}{d\theta} \ln g(\theta)^k \theta^{x-1} a(x) \left\{\frac{g(\theta)x - \theta g'(\theta)}{g^2(\theta)}\right\}\right. \\
= & (-k) \left\{\theta \frac{d^2}{d\theta^2} \ln g(\theta) + \frac{d}{d\theta} \ln g(\theta)\right\} \mu_{k-1} + \frac{1}{\theta} \Sigma (x - E(X))^k \frac{\theta^x (a(x))}{g(\theta)} \left\{x - \frac{\theta g'(\theta)}{g(\theta)}\right\} \\
\frac{d}{d\theta}\mu_k = & (-k) \left\{\theta \frac{d^2}{d\theta^2} \ln g(\theta)^2 + \frac{d}{d\theta} \ln g(\theta)\right\} \mu_{k-1} + \frac{1}{\theta} \mu_{k+1}
\end{aligned}
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